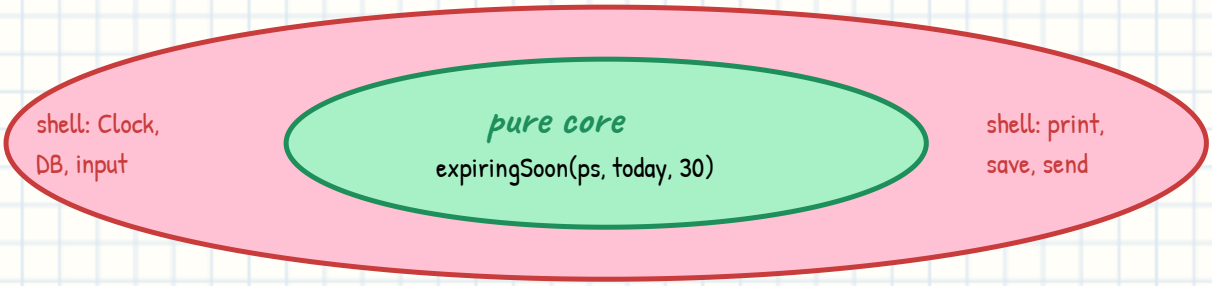
FP principles · my notes L15

small, pure, composable pieces

Functional core, imperative shell 🥚



→ POL-509 in 5 days, POL-505 in 13, POL-502 in 23, POL-508 in 28 · same with Clock.fixed ✓

Shallow records 🪞

record Statement
txnIds → caller's list!

[T009, T012, T999]

fix: List.copyOf(..)
→ [T009, T012]

unmodifiableList = a window onto a mutable list: view=[A, B], copy=[A]

Curry 🍛

`rate -> amount -> fee`

fix the rate → standardFee
1432.93 → 3.58 · premium → 1.43

Factories 🏭

`percentOff(10)` → a function
1240.50 → 1116.45
3 times → 904.33

Compose 🔗

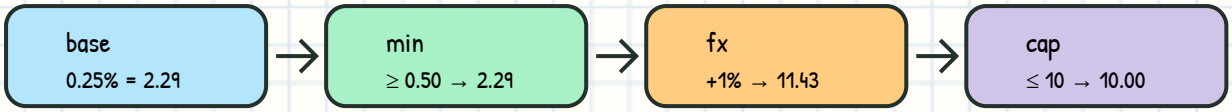
andThen: (1+2)×10 = 30
compose: 1×10+2 = 12
reduce(identity(), andThen)

Lazy 😴

orElse → evaluated 1×
orElseGet → 0×
Supplier log message: built only when on

Rules as a pipeline 🧪

T023 914.46 USD



record FeeRule(name, step) · List of rules · fold over an immutable Quote → fee + audit trail

Worked example: fee calculator

BEFORE: one class, fused if/else, hidden state · AFTER [8]: BASE.andThen(MINIMUM).andThen(FX).andThen(CAP) · MODERN: rules as data + audit · {SGD=42.95, USD=16.66} on 45 debits · all 3 agree ✓

Memoize: the trap 🪤

12 FX calls → 2 real lookups ✓ · recursive `computeIfAbsent` → ConcurrentModificationException ✕ · fix: get, compute, put → fib(90) instantly

Exercise: renewal quotes 📄

tierDiscount(tier) → when(> 500, −25) → +9% GST · POL-502 GOLD: 1240.50 → 1257.29 · POL-509 SILVER: 210.00 → 228.90

★ Remember

pure core, I/O at the edges · copy collections in records · factories, curry, partial · compose with identity · lazy Supplier · memoize pure only · next → L16 records, sealed types, patterns